

THULANI MTSHWELO

Software Engineer | Full-Stack Developer | Platform Architect

Cape Town, South Africa | [Email](#) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [AutoPromote](#)

PROFESSIONAL SUMMARY

Software engineer and full-stack platform builder experienced in designing and deploying enterprise and product-focused systems using JavaScript, TypeScript, Python, React, Next.js, Node.js and Django. Employed by Fine Chemicals Corporation while delivering substantial enterprise software work supporting Aspen Pharmacare Holdings Limited. Experienced in Azure AD/Entra ID, Single Sign-On, SharePoint, Power Platform, API integrations, cloud infrastructure and automated media-processing workflows. Founder and technical architect of AutoPromote, an automated multi-camera podcast editing and content intelligence platform.

CORE TECHNICAL SKILLS

Languages & Frontend

JavaScript, TypeScript/TSX, Python, React, Vue, Next.js

Backend & Data

Node.js, Django, REST APIs, Firebase, Supabase, PostgreSQL

Enterprise & Cloud

Azure AD/Entra ID, SSO, SharePoint, Power Apps, Power Automate, Power BI, CI/CD, cloud deployment

AI & Media Systems

FFmpeg, OpenCV, Pillow, Librosa, AI integrations, video/audio processing, workflow automation

PROFESSIONAL EXPERIENCE

Full-Stack Developer — Fine Chemicals Corporation | Nov 2025 – Present

Enterprise software delivery supporting Aspen Pharmacare Holdings Limited

Develop enterprise software and workflow solutions for internal business operations, including systems supporting Aspen Pharmacare Holdings Limited.

Developed the Integrity Hub platform for conflict-of-interest declaration management and related governance workflows.

Integrated Azure Active Directory/Entra ID authentication and Single Sign-On to support secure enterprise access.

Built internal tools and reporting solutions using SharePoint, Power Apps, Power Automate and Power BI.

Designed backend APIs, frontend interfaces and data workflows for enterprise systems.

Founder & Technical Architect — AutoPromote | Independent Product

Designed and built an automated multi-camera podcast editing and content intelligence platform from concept through working MVP.

Architected video and audio processing workflows using FFmpeg, OpenCV, Pillow and Librosa, including synchronisation, camera switching and media conversion.

Integrated publishing and platform APIs across TikTok, YouTube, LinkedIn, Reddit and Meta.

Implemented Firebase-backed workflows, cloud deployment, compression, CDN delivery and autoscaling strategies to control infrastructure cost.

Secured Meta approval for platform integrations and managed the technical approval process.

SELECTED ENGINEERING WORK

Integrity Hub — Enterprise declaration-management platform with Azure AD/Entra ID SSO, governance workflows and internal reporting.

AutoPromote — Automated multi-camera podcast editing platform combining video processing, AI-assisted workflows, cloud infrastructure and social-platform integrations.

Lean Product Architecture — Built scalable prototypes using Firebase, open-source tooling and cost-conscious cloud design.

EDUCATION

NQF Level 5 Software Development — Siseko Technologies, 2023–2024

National Senior Certificate — Isikhoba Nombewu Technical School, 2021

References available upon request